COURSE NAME

OBJECT ORIENTED ANALYSIS AND DESIGN

CSC 2210

(UNDERGRADUATE)





CHAPTER 2

USE CASE DIAGRAM

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USAGE OF USE CASE DIAGRAM?

- Capture the intended behavior of the system
- Without specifying how that behavior is implemented.
- Use cases provide a way for developers to come to a common understanding with your system's end users and domain experts.
- In addition, use cases serve to help validate your architecture and to verify your system as it evolves during development.
- Use cases specify **WHAT** are the externally visible behaviors; they **DO NOT** dictate **HOW** that behavior will be carried out internally.

USAGE OF USE CASE DIAGRAM?

- Use case diagrams are used to visualize, specify, construct, and document the (intended) behavior of the system, during requirements capture and analysis.
- Provide a way for developers, domain experts, and end-users to Communicate.
- Serve as basis for testing.
- Use case diagrams contain use cases, actors, and their relationships.

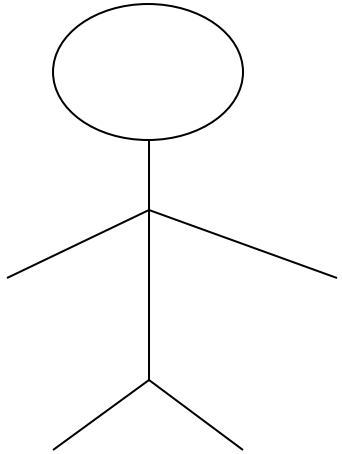
USE CASE

- Use cases specify desired behavior.
- A use case is a description of a set of sequences of actions, including variants (alternatives), a system performs to yield an observable result of value to an actor.
- For example, one central use case of a bank is to process loans for its loan applicants/ clients/ customers.
- The names of use cases are always written in the form of a verb followed by an object.
- Each sequence represent an interaction of actors with the system.
- Describing the flow of events within the use case.
- Can be done in natural language, formal language or pseudo-code.



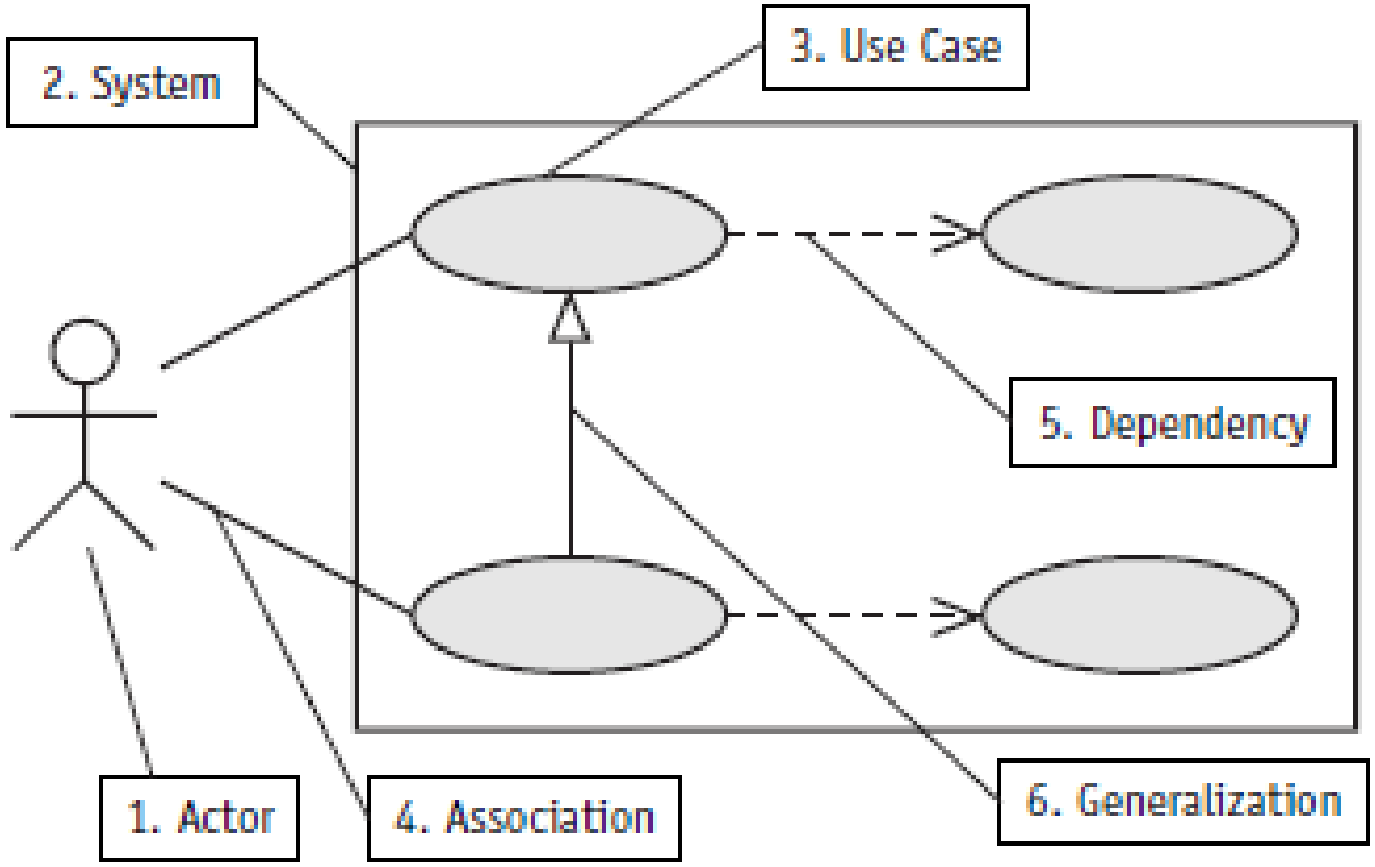
ACTOR

- An actor represents a set of roles that users of use case play when interacting with these use cases.
- Actors can be human or automated systems.
- For example, in modeling a bank, processing a loan involves, among other things, the interaction between a customer and a loan officer.
- Actors are entities
  - which require help from the system to perform their task, or
  - are needed to execute the system's functions.
- Actors are not part of the system.
- A system can be an Actor of other systems



Actor

ELEMENTS OF USE CASE DIAGRAM



RELATIONSHIPS IN USE CASE DIAGRAM

- **Actor – to - Use case**

  Association

  - Between Actor and its Use Cases
  - **NOT** between Use Cases
  - **NOT** between Actors

- **Use case – to - Use case**

  Generalization

  Dependency

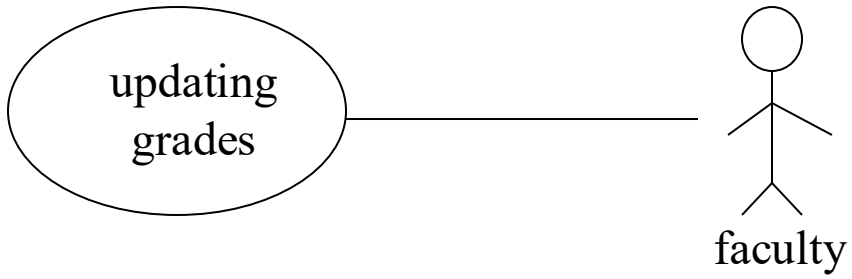
  - Between Use Cases ONLY
  - <<include>> and <<extend>>

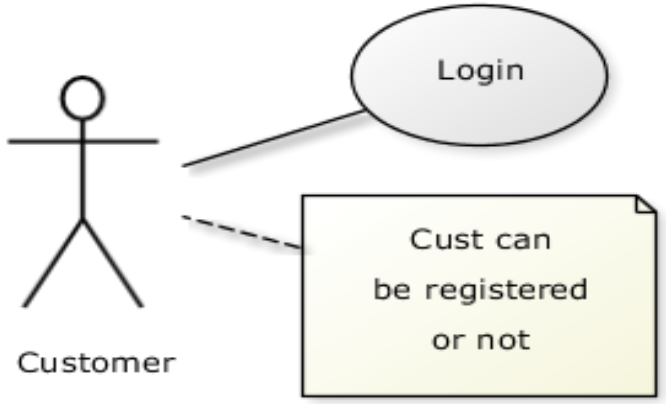
- **Actor – to - Actor**

  Generalization

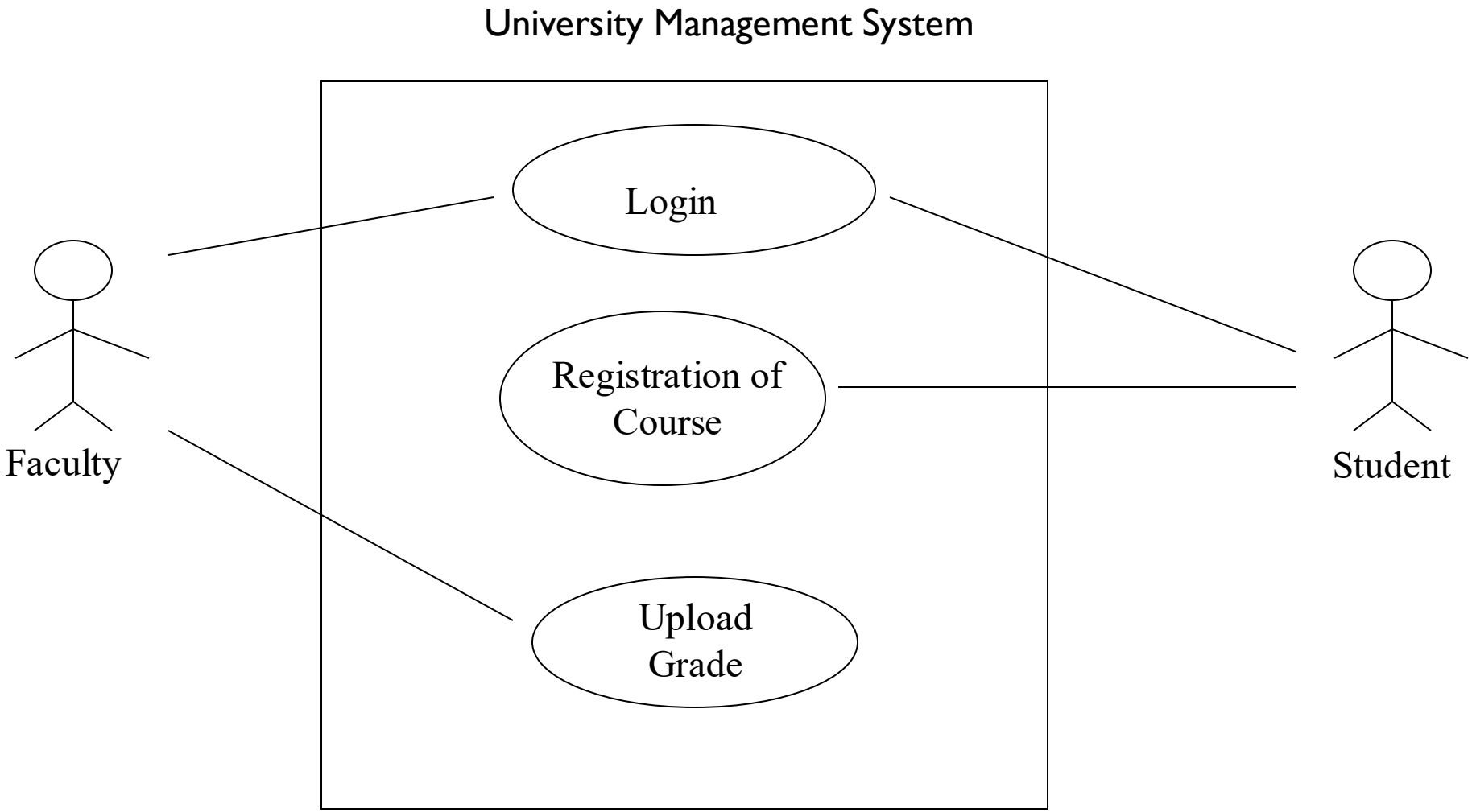
USE CASES AND ACTOR

- From the perspective of a given actor, a use case does something that is of value to the actor, such as calculate a result or change the state of an object.
- The Actors define the environments in which the system lives
- Actors may be connected to use cases by **associations**, indicating that the actor and the use case communicate with one another using messages.



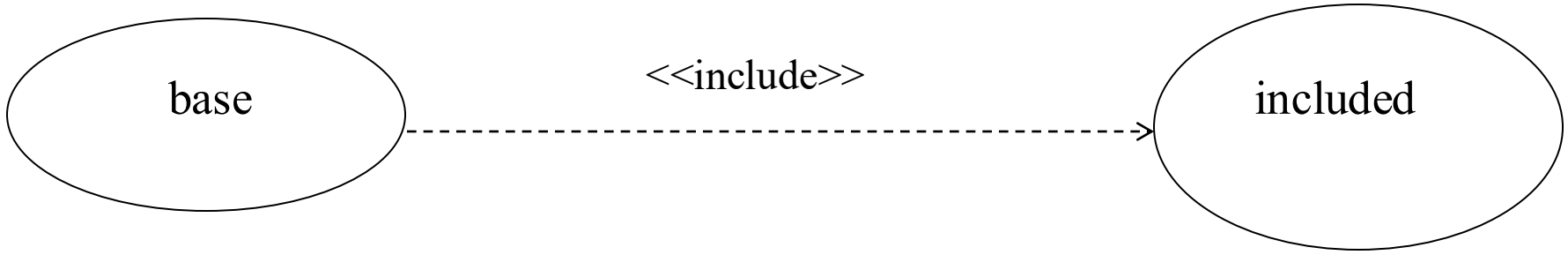


EXAMPLE OF USE CASES DIAGRAM



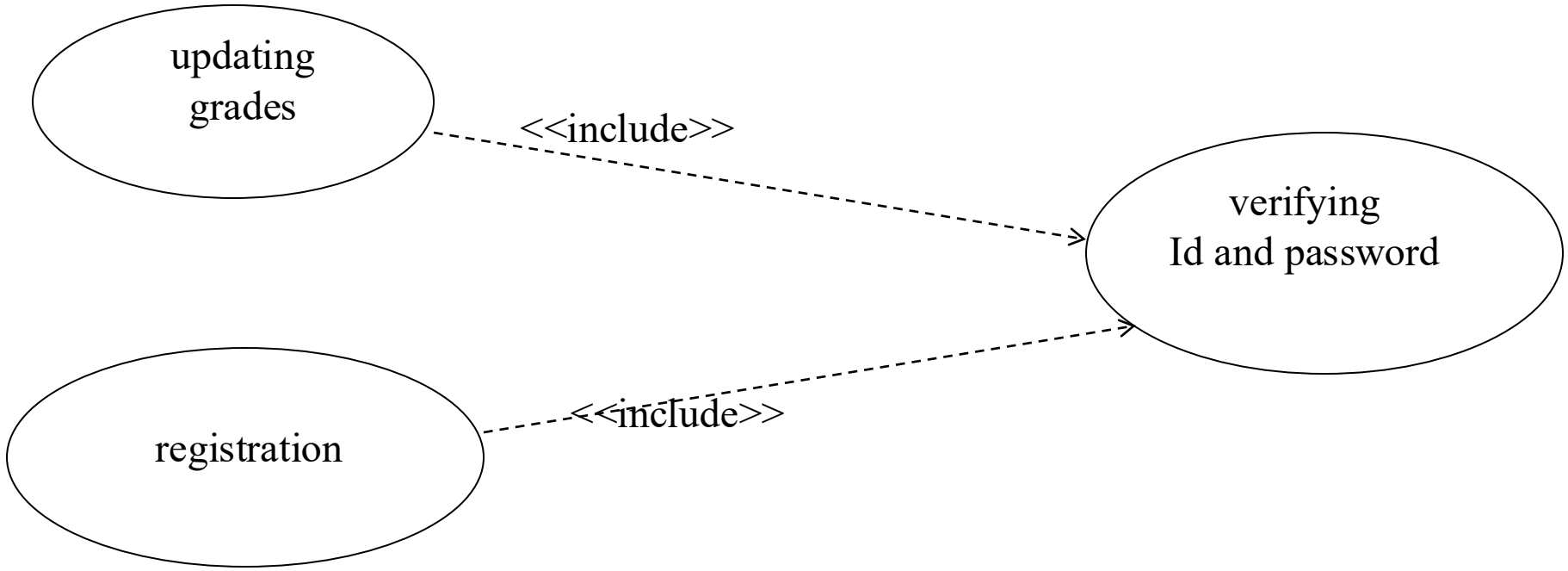
RELATIONSHIPS BETWEEN USE CASES

- **Include** - use cases that are included as parts of other use. Enable to factor common behavior
  - The base use case explicitly incorporates the behavior of another use case at a location specified in the base.
  - The included use case never stands alone. It only occurs as a part of some larger base that includes it. The original use case is not complete without the included one



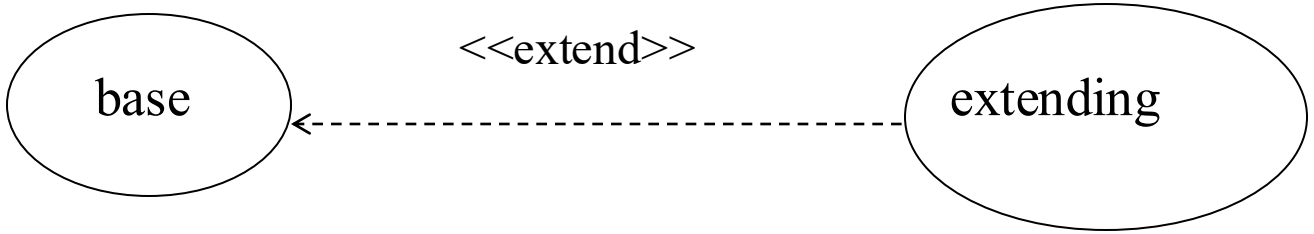
RELATIONSHIPS BETWEEN USE CASES

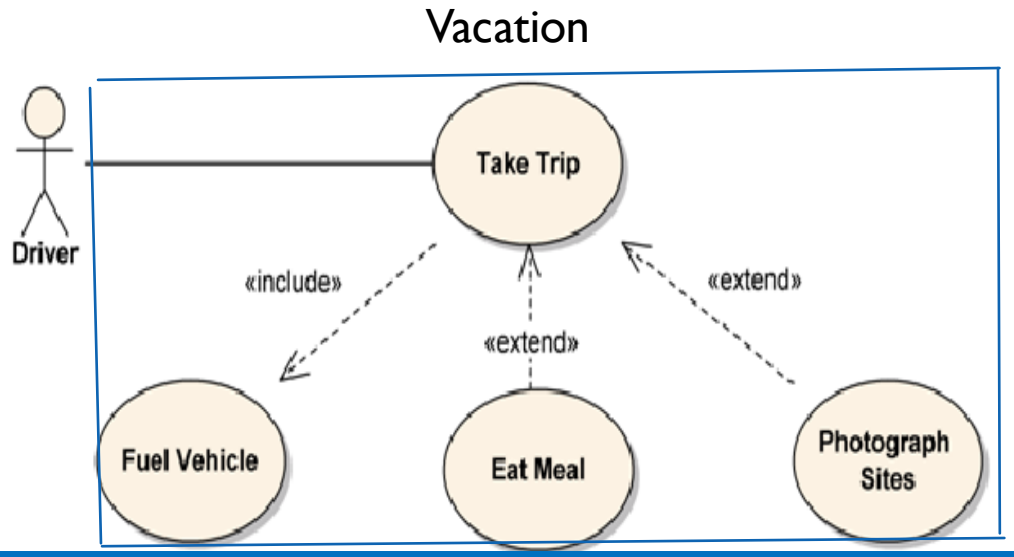
- Enables to avoid describing the same flow of events several times by putting the common behavior in a use case of its own.



RELATIONSHIPS BETWEEN USE CASES

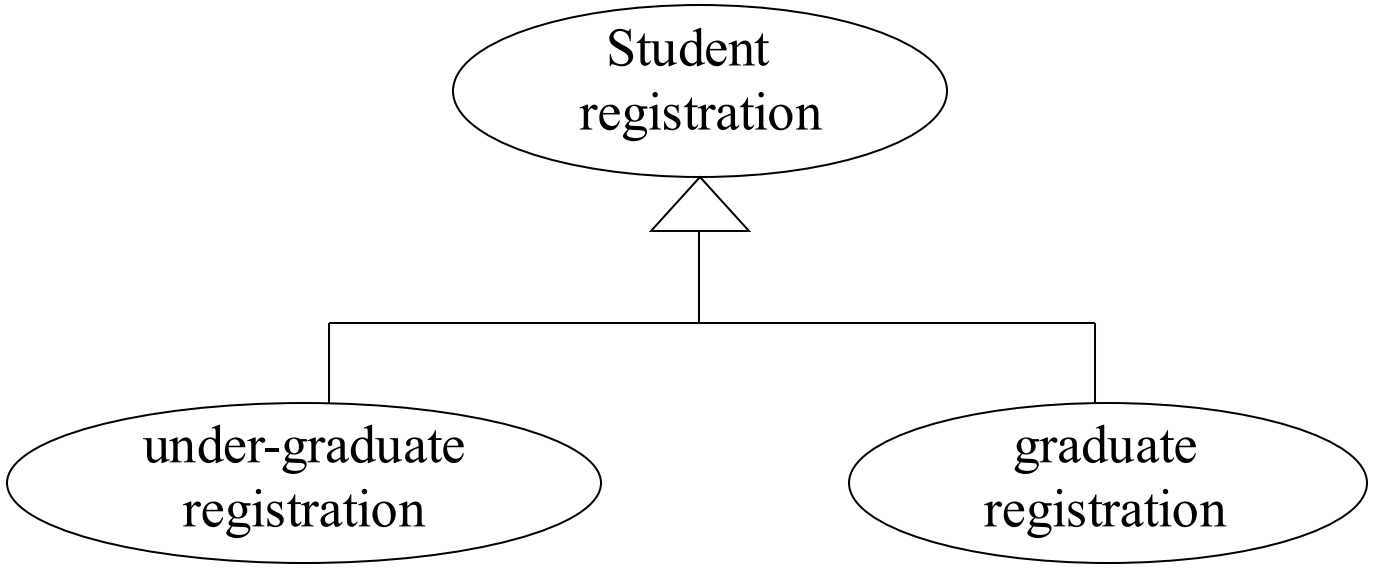
- **Extend** – a part of a use case that is *optional* system behavior
  - The base use case implicitly incorporates the behavior of another use case at certain points called extension points.
  - The base use case may stand alone, but under certain conditions its behavior may be extended by the behavior of another use case.
  - Enables to model optional behavior or branching under conditions.





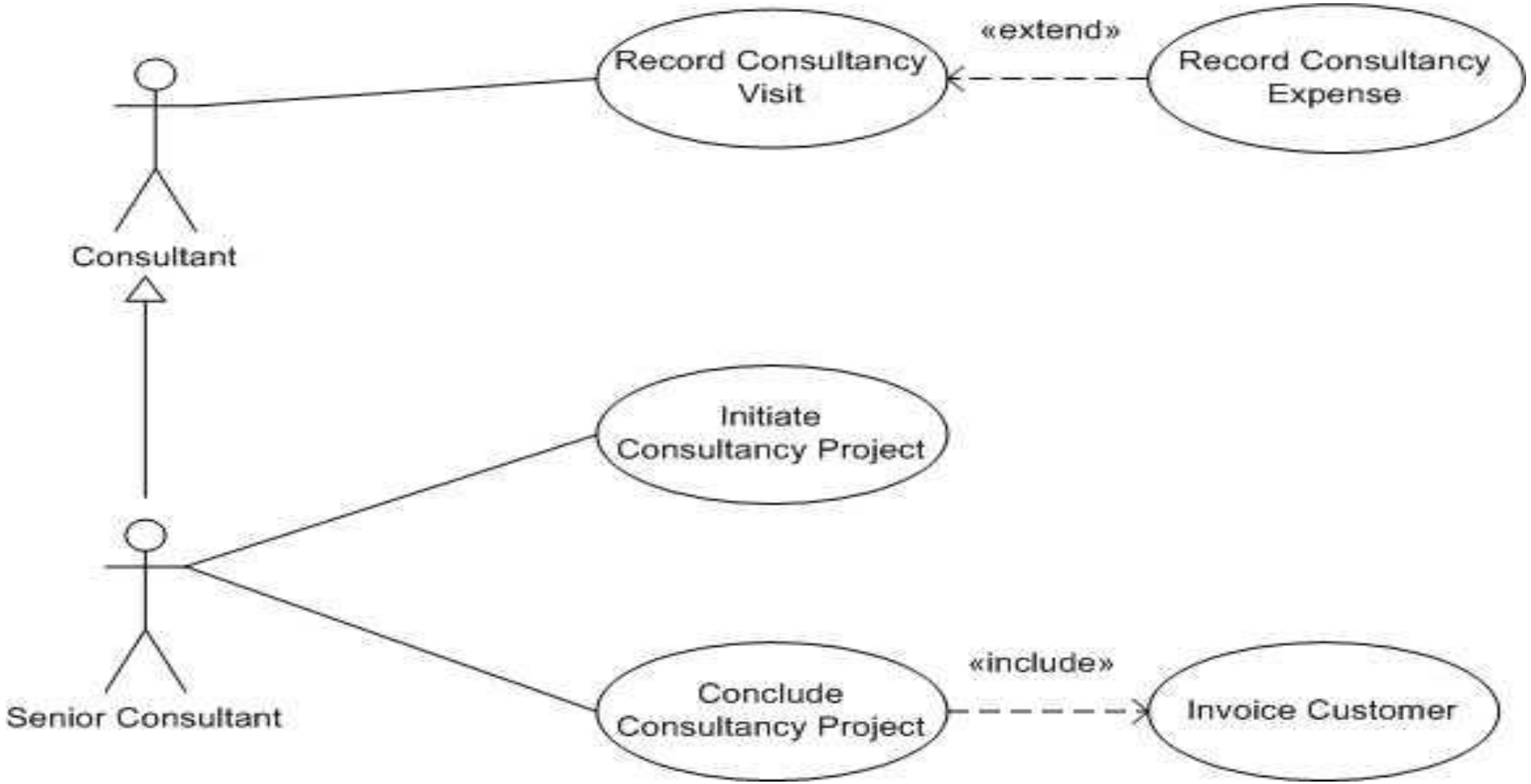
RELATIONSHIPS BETWEEN USE CASES

- **Generalization** - A generalization relationship means that a child use case inherits the behavior and meaning of the parent use case. The child may add or override the behavior of the parent.



RELATIONSHIPS BETWEEN ACTORS

- **Generalization** - A generalization relationship means that a child actor inherits the behavior and meaning of the parent actor. The child may add or override the behavior of the parent.



CASE STUDIES: USE CASE DIAGRAM

❑ Case 1:

In a hotel management system, a guest can rent rooms. Hotel receptionist uses the system to assist in the renting. A guest can also book a room for future renting with the help of the receptionist, but he has to check if the room has prior booking or not. A Guest pays for the rooms at the time they check out. He can pay by cash, cheque or credit card. Receptionists has to logon to the system before they can use it, but to logon he has go through username/password verification.

CASE STUDIES: USE CASE DIAGRAM

❑ Case 2:

Music Today is an audio recording company. There are several recordists in the company. Clients record their music with the help of the recordists. The recording is done in a session. A session is usually created by the office clerk. A session needs to be booked by the client beforehand and before booking the time needs to be verified by the booking clerk for availability. The company maintains a list of preferred clients who book sessions like regular clients but also can create sessions according to their needs. Clerks have to log in to use the system. Clients can pay in cash or card. The accountant deals with the payment. The payment is only cleared when the accountant receives clearance from the booking clerk.

CASE STUDIES: USE CASE DIAGRAM

❑ Case 3:

In a hospital management system, a patient's medical history is created by a doctor. A patient may be referred by a doctor to be admitted in the hospital. A patient can rent hospital facilities. An administrative officer deals with the renting. The system automatically checks whether the patient is referred by a doctor before he can rent any hospital facility. The types of facilities are rooms, beds or ICUs. Admitted patients are regularly visited by doctors and a nurse updates patient medical history after each visit. A doctor writes the discharge note of the patient when he leaves which is also included in the patient's medical history. An account clerk prepares the bill. The bill is calculated from the elements written in the medical history, i.e. number of doctor's visit, prescribed medication, tests. When the nurse updates the medical history she writes either the date-time of doctor's visit or prescribed medication or test. The patient may pay by cash or card when the bill is prepared.

CASE STUDIES: USE CASE DIAGRAM

❑ Case 4:

ABCD Records is a mail order company that distributes CDs and tapes at discount prices to record club members. A member fills up an order form and sends it to the order processing clerk. When the order processing clerk receives an order form, she verifies that the sender is a club member by checking the member file. She also checks the status of the member whether he is a royal class member or regular class member. If the sender is not a member, the clerk returns the order along with a membership application form. If the customer is a member, the clerk verifies the order item data by checking the item file. Then the clerk enters the order data and saves it to the daily records file. At the same time the clerk also prints an invoice and shipping list for each order, which are forwarded to the collection department clerk for processing there. If the items are not available they are ordered if the member is a royal class member. Royal class members also get discounts. The members can pay by cash, check or bank draft.

REFERENCES

- Booch, G., Rumbaugh, J. & Jacobson, I. (2005). *The unified modeling language user guide*. Pearson Education India.